

Shreeya Dasa Lakshminath

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ML engineer with production LLM experience and an M.S. in Applied Machine Learning. Builds and deploys LLM systems end-to-end: fine-tuning, RAG, retrieval-ranking, and inference, with research-grade fundamentals in RL and probabilistic ML.

EXPERIENCE

Juniper Networks

Software Engineer 2

Bangalore, India

Feb 2024 - Jul 2025

- Built an LLM-powered network-log chatbot: ingested real-time router telemetry via Kafka, fine-tuned a pretrained LLM on domain-specific log data, and layered a RAG pipeline over live router state, enabling natural-language debugging and eliminating manual log parsing for engineers.
- Developed an AI-driven test-script generation tool that converted natural-language test specifications into executable validation scripts for SD-WAN network configurations, cutting manual script-writing effort and accelerating release-cycle turnaround across the team.

Vector NDT Technologies

Contract ML Engineer

Bangalore, India

May 2023 - Oct 2023

- Built a YOLOv8 computer-vision pipeline for automated crack detection on engine parts, achieving 95% accuracy on the client validation set and eliminating manual visual inspection for in-scope part classes; owned data labelling, augmentation, training, and inference.

BMS College of Engineering - Research Collaboration with IISc & St. John's Hospital

Undergraduate Research Intern

Bangalore, India

Jan 2022 - Jun 2023

- Collaborated with researchers at the Indian Institute of Science (IISc) and St. John's Hospital on assistive technology for early dyslexia diagnosis from EEG signals; built the Android-side acquisition and visualization layer for real-time EEG streams used in the clinical data collection workflow.

PROJECTS

MCP Auditor: Multi-Agent Security Auditor for AI Agent Infrastructure | TypeScript, AWS Bedrock, Next.js, Render | [LiveDemo](#)

- Built and deployed live (Render) a multi-agent system that audits MCP servers for security flaws: six prober agents run in parallel across distinct vulnerability classes (prompt injection, credential leakage, scope violation, auth bypass), each gated by a governance and audit layer.
- Synthesized probe evidence into a cited security report grounded in the SAFE-T taxonomy, auto-filed as GitHub issues, behind an HTTP 402 payment gate; built end-to-end in a 1-day hackathon.

Autonomous Enterprise RAG Agent with Human-in-the-Loop Feedback | LangGraph, FastAPI, Streamlit | [GitHub](#)

- Built a multi-agent RAG system with LangGraph that decomposes queries, routes them across retrievers, and synthesises structured responses; designed a HITL feedback pipeline that captures preference pairs formatted for a downstream DPO/SFT fine-tuning loop.

LLM-to-Cross-Encoder Distillation for Retrieval Reranking | PyTorch, Hugging Face Transformers | [GitHub](#)

- Fine-tuned LLaMA 3 with LoRA as a cross-encoder for retrieval ranking in RAG, and applied knowledge distillation from a larger teacher to compress the model while preserving relevance, improving reranking quality by +15% MRR and +12% NDCG over the base retriever.
- Work written up as a co-authored manuscript, "Transforming LLMs into Efficient Cross-Encoders," in preparation, 2026.

GPT-2 (124M) Pretrained From Scratch on a Statistical-Domain Corpus | PyTorch, Python | [GitHub](#)

- Trained a 124M-parameter GPT-2 from scratch on a curated statistical-textbook corpus, reaching 4.82 cross-entropy (perplexity ~124) after 3 epochs, a ~40% perplexity reduction vs. the same architecture initialized at uniform-token loss.
- End-to-end pretraining stack: PDF extraction, deduplication, BPE tokenization, sharded data loading, training loop with gradient accumulation and mixed-precision training, and sample generation for qualitative evaluation.

Nonparametric Bayesian IRL with Data-Parallel Gibbs Sampling | JAX, Ray, Python, UMD Zaratan HPC | [GitHub](#)

- Designed a Dirichlet Process mixture over reward functions to jointly infer the number of latent expert types and their rewards from 20 multi-expert trajectories, achieving ARI = 1.000 with no prior knowledge of K.
- Built a collapsed Gibbs sampler combining Chinese Restaurant Process cluster assignments with Metropolis-Hastings reward-weight updates; all likelihoods evaluated strictly in log-space to prevent Boltzmann-policy underflow.
- Parallelized inference across up to 16 CPU workers via Ray on the UMD Zaratan HPC cluster, 4.79x peak speedup; characterized a throughput-accuracy tradeoff from the consensus merge heuristic.

Ridge, Lasso & Elastic Net via Convex Optimization (MSML 604) | Python, NumPy, Streamlit | [GitHub](#)

- Built ISTA/FISTA proximal-gradient solvers from scratch with Fenchel duality-gap stopping certificates; validated vs. scikit-learn (max error < 1e-2) on a 180K-record dataset, Lasso lowest RMSE (29.96) with 16/39 active features.

EDUCATION

University of Maryland, College Park

M.S., Applied Machine Learning, GPA: 3.9/4.0

College Park, MD

Aug 2025 - May 2027 (Expected)

BMS College of Engineering

B.E., Computer Science & Engineering, CGPA: 8.28/10

Bangalore, India

Aug 2020 - Jun 2024

SKILLS

Languages: Python, SQL, C++, Java

ML / Deep Learning: PyTorch, JAX, Hugging Face (Transformers, PEFT/LoRA, TRL, Datasets, Accelerate), W&B, scikit-learn, NumPy, Pandas

LLMs, RL & Probabilistic ML: LLM pretraining/post-training (SFT, LoRA, DPO), reranking, knowledge distillation, RAG, LangGraph; IRL, MDPs, Soft Value Iteration, Dirichlet Process, Gibbs/MH

Infra & Deployment: Docker, AWS, GitHub Actions (CI), Ray, SLURM/HPC, FastAPI, PostgreSQL, Linux, Git, Jenkins